Renwei Meng

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EDUCATION

Anhui University, Hefei, China — Sep. 2023 – Jun. 2027

B.Eng. in Digital Media Technology — **GPA: 3.95/4.0**; 4.34/5.0; 92.58/100; **Rank: 1/100**; **IELTS: 7.0**

Core Coursework: Diffusion & Flow Matching, Reinforcement Learning, Machine Learning, Linear Algebra, Probability Theory, Data Structures, Computer Organization, Operating Systems, Computer Networks.

RESEARCH INTERESTS

Trustworthy Generative AI (manifold-aware evaluation, reward hacking, reinforcement learning, flow matching); **Reliable LLMs and Agentic AI** (knowledge boundaries, abstention, RAG, multi-agent auditing); **Multimodal and Neurocognitive Intelligence** (EEG decoding, BCI, gaze modeling, synthetic-media detection).

PUBLICATIONS & PATENTS

Publications — * Equal contribution

- **Normal Fréchet Drift: A Geometric Measure for Detecting Off-Manifold Deviation in Generative Models**. **Renwei Meng***, Yansen Han*, Tao Lin. *ICLR 2027.* **Under Review**.
  Introduces a geometry-aware metric for quantifying harmful normal drift from pretrained data manifolds.
- **MORPH: Bi-Temporal Online Reinforcement Learning for Structure-Aware Music Flow Matching**. **Renwei Meng**. *AAAI 2027.* **Under Review**.
  Performs counterfactual credit assignment jointly over flow steps and musical sections for long-form generation.
- **Know2Guess: A Contamination-Aware Multi-Zone Benchmark for Knowledge-Boundary Evaluation in Large Language Models**. **Renwei Meng**, Bowen Zhang, Jian Wang, et al. *ICONIP 2026.* **Accepted**.
  Builds a 1,200-item, four-zone benchmark for answerability, abstention, refusal, and contamination.
- **Conditional Riemannian Adaptive Flow Transport for Cross-Subject EEG Decoding**. **Renwei Meng**, Hongyi Zhang. *PRICAI 2026.* **Accepted**.
  Combines SPD alignment, neural residual flows, and label-free test-time adaptation across five benchmarks.
- **Group Resonance Network (GRN) for Cross-Subject EEG Emotion Recognition**. **Renwei Meng**. *ICANN 2026.* **Accepted**.
  Integrates learnable group prototypes with PLV/coherence-based multi-subject resonance modeling.
- **S-AODP++: Structure-Aware AOI-Guided Scanpath Attention for AI-Generated Image Identification**. Jingming Wang*, **Renwei Meng***, Yunjie Si*. *ICMI 2026.* **Accepted**.
  Uses AOI transition graphs and scanpath attention, achieving 81.7% accuracy and 85.0% AUROC.
- **Explainable Innovation Engine: Dual-Tree Agent-RAG with Methods-as-Nodes and Verifiable Write-Back**. **Renwei Meng**. *arXiv preprint.*
  Organizes methods in provenance and abstraction trees for controllable synthesis and verified write-back.
- **Breaking the Support Barrier in RLVR: A Fork-Conditional Reachability Criterion for Reliable Boundary Expansion**. **Renwei Meng**. *ACML 2026.* **Under Review**.
  Introduces FCR-PO for predictable, verifier-gated support expansion at substantially lower compute than tree search.
- **VulnAgent-R2: Evidence-Calibrated Multi-Agent Auditing for Repository-Level Vulnerability Detection**. **Renwei Meng***, Haoyi Wu*, Jingming Wang*. *ACML 2026.* **Under Review**.
  Integrates risk screening, repository context expansion, specialist agents, and evidence fusion.

Patents

- **A generative deepfake video detection method and its system**. Shiang Hu, Zhiwen Zha, **Renwei Meng**, Yuhao Fang. China Invention Patent No. 202610165744.1. **Accepted**.
- **Resting-state EEG quality assessment based on dual-branch contrastive learning**. Shiang Hu, Dongdong Jia, Wan'er Lv, **Renwei Meng**, Chao Lv. China Invention Patent No. 202512044739.3. **Accepted**.

RESEARCH EXPERIENCE

Manifold-Aware Detection of Reward Hacking in Generative Models — Mar. 2026 – Present

Core Researcher | Advisor: Prof. Tao Lin | Westlake University

- Proposed **Normal Fréchet Drift**, which reconstructs local data geometry and separates tangential variation from harmful off-manifold drift, with theoretical guarantees and oracle-free hyperparameter selection.
- Validated NFD on analytical manifolds and text-to-image checkpoints; it detected degradation missed by reward, FID, nearest-neighbor, and divergence-based baselines. Manuscript under review at **ICLR 2027**.

Inter-Brain Synchrony for AI-Generated Content Detection Sep. 2025 – Sep. 2026

Core Researcher | Advisor: Prof. Shiang Hu

- Investigated AI-generated media detection through stimulus-locked cross-subject neural responses using dual-brain EEG hyperscanning experiments.
- Built a neural-cognition-inspired framework using multi-band features, Transformers, GNNs, group prototypes, synchrony features, and Riemannian alignment.
- Achieved **93.7% accuracy** and **92.9% F1-score**, outperforming CNN, SVM, and standard multimodal baselines.

Industrial Metaverse Intelligent Interaction System Sep. 2025 – Jun. 2026

Algorithm Engineer | Advisor: Prof. Shiang Hu

- Built a real-time multimodal interaction system integrating eye tracking, gesture recognition, and pose estimation.
- Designed low-latency intent-fusion algorithms and optimized 3D rendering and data transmission for industrial scenarios.

Transformer-Based Software Vulnerability Detection Mar. 2025 – Jan. 2026

Project Leader | National College Student Innovation Program | Anhui University

- Developed Transformer-based vulnerability detection over AST and CFG representations to capture long-range code dependencies.
- Used self-attention to extract deep semantic features from program structure and long-range execution context.
- Curated a large-scale C/C++ dataset and evaluated precision and recall against static-analysis tools, leading to VulnAgent-R2.

Selected Earlier Projects

Coronary CTA Artifact Removal Jan. 2025 – Jun. 2025

Research Assistant | Advisor: Prof. Zhe Jin | Anhui University

- Combined U-Net and GAN-based denoising to remove motion artifacts, improving SNR and SSIM over traditional filters.

Carbon Price Prediction Jun. 2024 – Jun. 2025

Project Leader | National College Student Innovation Program | Anhui University

- Led development of a CNN-GRU-LSTM-Attention model that improved prediction accuracy across seven Chinese carbon markets.

INTERNSHIP EXPERIENCE

Taiyuan Longwei Electronic Technology Co., Ltd. — Java Backend Developer Intern Jul. 2025 – Sep. 2025

- Built Spring Boot/MyBatis services and RESTful APIs; improved data transmission efficiency and backend stability through frontend-backend collaboration.
- Managed database operations and coordinated interface design with the frontend team for stable service integration.

State Grid Shanxi Electric Power Co., Ltd. — Systems Operations Intern Jul. 2024 – Sep. 2024

- Maintained critical power-system services and analyzed operational metrics for monthly optimization and reliability reports.
- Supported platform availability and security for critical grid-management workflows.

HONORS & AWARDS

- **National Second Prize**, National College Students Market Survey and Analysis Competition 2026
- **Finalist**, Mathematical Contest in Modeling (MCM/ICM) 2026
- **National Scholarship** (Top 0.1%, highest undergraduate honor in China) 2025
- **First Prize**, National College Student Mathematical Modeling Competition (Provincial) 2025
- **Second Prize**, Lanqiao Cup National Software Talent Competition (Java, Provincial) 2025
- **Honorable Mention**, Mathematical Contest in Modeling (MCM/ICM) 2025
- **University First-Class Academic Scholarship** 2024, 2025

TECHNICAL SKILLS

Programming: Python, MATLAB, C/C++, Java, R, SQL, HTML, Bash, zsh.

Frameworks & Tools: PyTorch, TensorFlow, Git, Docker, Linux, E-Prime 3, LangChain, LangGraph.

Additional: LaTeX, PowerPoint, Figma, Visio, Adobe Photoshop/Illustrator, Blender (3D modeling); ChatGPT, Codex, Claude Code, n8n, ComfyUI.